

Vision–Language Learning for Remote Sensing: From Change Detection to Super-Resolution

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Abstract—This summary presents two advances in multimodal learning for remote sensing: **Multimodal Difference Augmentation for Change Detection (MdaCD)** and **Text-Conditioned Remote Sensing Super-Resolution (Text-RSSR)**. Both integrate vision–language models such as CLIP to bridge the semantic gap in traditional remote sensing analysis. Papers of MdaCD and Text-RSSR can be found in Hao Yang’s personal website.

Index Terms—Remote sensing, remote sensing change detection, remote sensing super-resolution, multimodal learning

I. MULTIMODAL DIFFERENCE AUGMENTATION FOR CHANGE DETECTION

FIGURE 1 illustrates the core motivation of MdaCD: in remote sensing change detection, textual descriptions generated by CLIP often emphasize unchanged regions while overlooking subtle but meaningful changes. This occurs because unchanged areas dominate the image and domain shifts distort textual alignment. As shown, CLIP’s descriptions misinterpret scene variations, highlighting harbors instead of industrial growth. MdaCD addresses this by guiding CLIP’s attention toward changed regions through CLIP-Guided Masking, enabling more accurate multimodal alignment and differential feature learning.

MdaCD introduces a CLIP-Guided Masking mechanism and a Multimodal Fusion and Difference Augmentation process to enhance change detection. The masking guides textual attention toward changed regions, while the fusion-augmentation module amplifies differential features across visual and textual modalities, achieving robust multimodal alignment and state-of-the-art performance in remote sensing change detection.

II. TEXT-CONDITIONED REMOTE SENSING SUPER-RESOLUTION

Figure 2 highlights the motivation of Text-RSSR: during satellite imaging, transmitting full high-resolution data to

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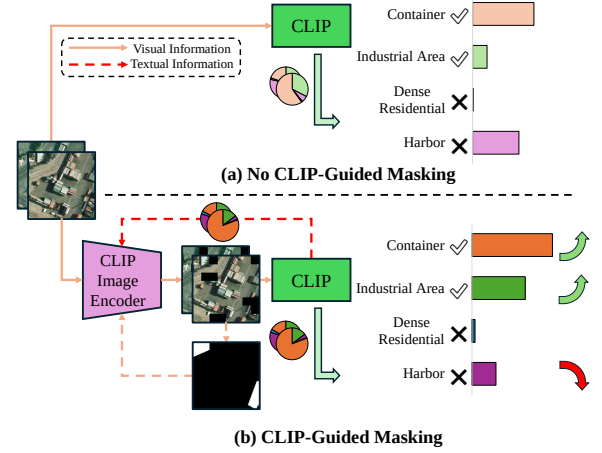


Fig. 1. The textual descriptions generated by the CLIP model, along with the corresponding original and masked bi-temporal image pairs, demonstrate that changes in the images can lead to unexplainable impacts in the generated descriptions.

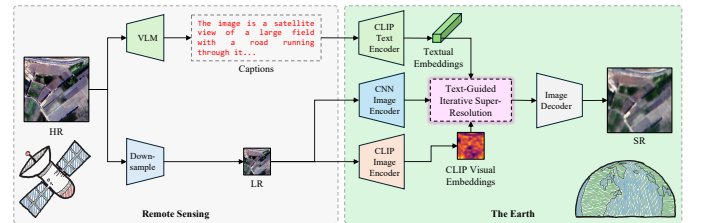


Fig. 2. Overall architecture of the proposed Text-RSSR.

Earth is costly, while low-resolution images lose essential high-frequency details. Text descriptions, however, require negligible storage and can preserve semantic and structural information of the original high-resolution scene before down-sampling. By integrating these captions as auxiliary inputs, Text-RSSR leverages textual semantics to guide visual reconstruction, enabling the recovery of fine spatial details and improving the fidelity of remote sensing super-resolution.

Text-RSSR introduces a novel Text-Guided Iterative Super-Resolution framework that aligns CNN visual embeddings with CLIP-based textual and visual embeddings through hierarchical feature refinement. This design enables cross-modal supervision, allowing semantic cues from text to guide fine-grained reconstruction, thus enhancing structural detail, contextual consistency, and domain adaptability in remote sensing imagery.